

#ASLI ENGINEERING

Eviction

SWIPE

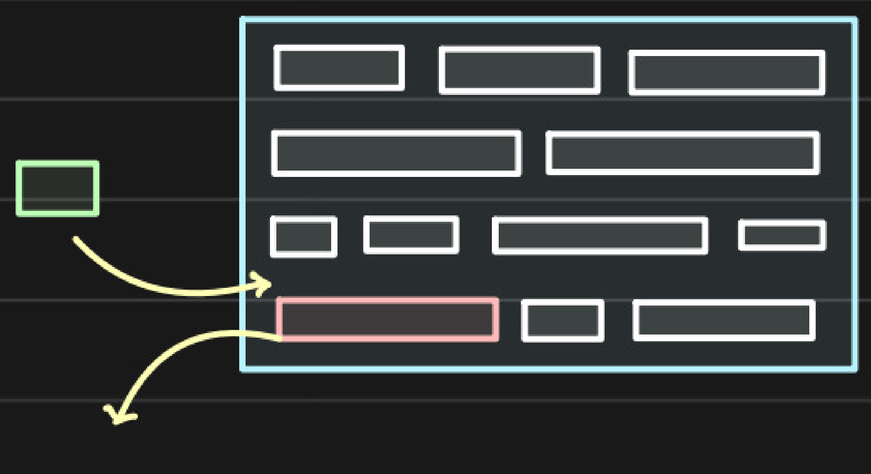
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Key Eviction

Redis is RAM-bound and we cannot have an unbounded-storage to accomodate any number of keys

Hence, to accomodate newer data,
we have to evict some old data.



How Redis Evicts?

Redis has a maxmemory configuration
that limits the max amount of data it could hold.

When the Redis data size hits the limit, it evicts some of the old data.
The data that would be evicted, depends
on the eviction strategy configured, some of them are

noeviction : new values aren't saved when memory limit is hit

allkeys-lru : removes least recently used keys

allkeys-lfu : removes least frequently used keys

volatile-lru : removes LRU keys with EXPIRE set

volatile-lfu : removes LFU keys with EXPIRE set

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allkeys-random : picks some keys at random and evicts them

volatile-random : picks some keys with EXPIRE set at random to evict

volatile-ttl : picks key with shortest TTL and evicts it

Approximated LRU [Redis 3.0]

Redis' LRU is not an exact algorithm. instead it is approximated

CORE Idea : Sample some keys
and from them evict the one that is least recently used

But why Redis does not use exact LRU ?

Because it requires extra memory

Imagine maintaining a DLL to maintain accesses - The additional memory overhead of maintaining pointers is an overkill

↑ Redis would choose to use that mem for data & not ↑

In approximated LRU,

we can take 'n' samples each having 'k' keys

and then evict the one which is least recently used

more samples, better accuracy to evict the actual LRU key

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